Statement of research interests

My research studies the economics of energy and labor market transitions. I'm keen on using advanced tools in macroeconomics to understand barriers to adoption and integration of new technologies, types of capital, and people. Specifically, my job market paper considers the impact of batteries on an optimal renewable energy transition, and how the electricity grid and US economy might respond to the introduction of clean energy aided by batteries. Another paper I'm working on considers the impact of immigration on inequality. This project seeks to model the mechanism of additional labor supply on demographic shifts and educational choices of natives. Lastly, I'm working on a project which models optimized and rule of thumb strategies for charging and discharging in a vehicle-to-grid integrated system for a fleet of electric vehicles. I will briefly describe these papers, and my pipeline for future work in this document.

My job market paper seeks to understand the role of batteries on an optimal renewable energy transition. Are they imperative to a renewable energy transition? Or is it better to invest in solar and wind? To what extent can we expect the electricity grid to migrate to renewable energy and batteries, unlocking electrification of the economy and higher demand for electricity? These questions are interesting and policy relevant for public debates regarding optimal taxation and subsidy responses to drive down emissions externalities. I develop a structural dynamic model for electricity generation deployments and investments, utilizing machine learning methods including approximating the value function with a neural network. I find that an optimal renewable energy transition drives renewable energy share of generation to 100% (aided by batteries), decreasing prices of electricity by 14% and overall emissions by 56%. I contrast these results with the amount of investment we might expect from a competitive equilibrium where electricity generators enter based on a profit motive. Finding a discrepancy here, I turn to an optimal policy response in the form of a carbon tax, closing this welfare gap with a tax of ~$100 / ton of CO_2.

A second project I'm working on also builds a model of the US economy, but instead focuses on another transition- this time of the labor market to one that incorporates immigrants. Immigrants enter the US with slightly different characteristics than natives, there is more variation in highly skilled and highly unskilled workers. This variation in skill level complements differently with capital, and other types of labor, to produce output. As such, we develop a production function to estimate the degree of complementarity between these four labor groups (skilled and unskilled, immigrant and native) and capital. We then (plan to) run a series of counterfactuals regarding demographic change, and immigrant intake policy to understand how immigration has and will continue to impact natives' educational choices, wages, and income inequality.

The last project I'm working on features a recently approved opportunity for electric vehicles to interact with the electricity grid. Utility scale batteries (in my job market paper) currently charge during the day, and discharge at night, taking advantage of an energy arbitrage to earn an intra-day profit. Electric vehicles can likewise take advantage of this mechanism, as well as enter the market for grid regulation services, or frequency balancing. This paper seeks to uncover an optimal charge and discharge decision for electric vehicles participating in these markets and compare it to a rule of thumb heuristic which seeks to explain this behavior.

Future projects that might be incorporated in this research agenda include extensions of my job market paper such as pairing this renewable energy transition with productivity increases, as evidenced by the ongoing data center rollout in the US increasing the demand for electricity, or adding spatial heterogeneity to the model to forecast where investments might take place. There is speculation that there may be complementarity between Europe and Africa in terms of solar potential, which might make for an interesting paper. Additionally, as climate change will impact geography differently, it might be interesting to model migration with a changing climate, as a means to combine these two projects that are objectively different topics but mathematically are similar in terms of how the models are solved.